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Late Ediacaran paleogeography of Avalonia and the Cambrian assembly of West Gondwana

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ABSTRACT

The latest Proterozoic tectonic evolution of Avalonia, one of the largest peri-Gondwanan terranes, bears on the issue of Gondwana amalgamation during the transition from Rodinia to Pangea. New paleomagnetic results from mid–late Ediacaran strata in the Newfoundland sector of Avalonia confirm substantial apparent motion of that terrane between ca. 590 and 560 Ma, consistent with a true polar wander interpretation for that interval. When the Avalonian poles are mapped onto complementary paleomagnetic records from West Africa and Laurentia, the Avalonia terrane may occupy any of four possible positions, all of which are far-removed from those cratons. Choosing the most kinematically parsimonious location leading toward Gondwana assembly, we find that Avalonia was most likely proximal to Amazonia during the Ediacaran–Cambrian transition, with a substantial (~3000 km) separation between Amazonia and West Africa at that time. Our results thus favor the existence of a wide Clymene Ocean and subsequent Cambrian final assembly of West Gondwana, providing a key snapshot onto the evolving Rodinia-Gondwana supercontinental transition.

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1. Introduction

The timing of orogenic assembly among Gondwana cratons is crucial to testing the existence of hypothetical supercontinent Pannotia (Murphy and Nance, 1991; Evans, 2020; Murphy et al., 2000) and thus evaluating the supercontinental cycle in general terms. Within West Gondwana, the most contentious debate on cratonic assembly concerns the timing of Amazonia's accretion to its Gondwana neighbors. Some researchers advocate that West Gondwana had welded together before ca. 600 Ma (e.g., Cordani et al., 2013); whilst others suggest that a substantial seaway (Clymene Ocean) between Amazonia and other Gondwana cratons was not completely closed until Cambrian time (e.g., Trindade et al., 2006; Tohver et al., 2010, 2012; McGee et al., 2015; Schmitt et al., 2018). Although recent key paleomagnetic poles from West Africa (Robert et al., 2017) show dramatic polar motions during mid–late Ediacaran time, the lack of definitive Ediacaran paleomagnetic poles from Amazonia leaves the controversy unresolved (D'Agrella-Filho et al., 2016).

Kinematic constraints from peri-Gondwanan terranes may assist in resolving this kind of controversy, if their motions can be linked plausibly to once-adjacent, larger cratons (i.e., to Amazonia). Avalonia is one of the largest peri-Gondwanan terranes, and its geological record provides important constrains on the tectonic history of the supercontinental transition from Rodinia to Pangea (e.g., Murphy et al., 2000, 2004, 2019; Pollock et al., 2012). Most current Rodinia reconstructions show Avalonia proximal to a nexus of West Africa, Amazonia, and Baltica (e.g., Pisarevsky et al., 2008; Li et al., 2013; Cawood et al., 2016). After Rodinia breakup (<750 Ma), different portions of Avalonia show distinct stratigraphic and geochronologic-isotopic affinities to these cratons, namely East Avalonia to West Africa (e.g., Nance and Murphy, 1994; Waldron et al., 2019), and West Avalonia to either Amazonia or Baltica (e.g., Pollock et al., 2009; Thompson et al., 2012; Henderson et al., 2016; Murphy et al., 2019). Herein, we present new Ediacaran paleomagnetic results from Avalonia in its type locality of Newfoundland, Canada (Fig. 1), and establish a defini-

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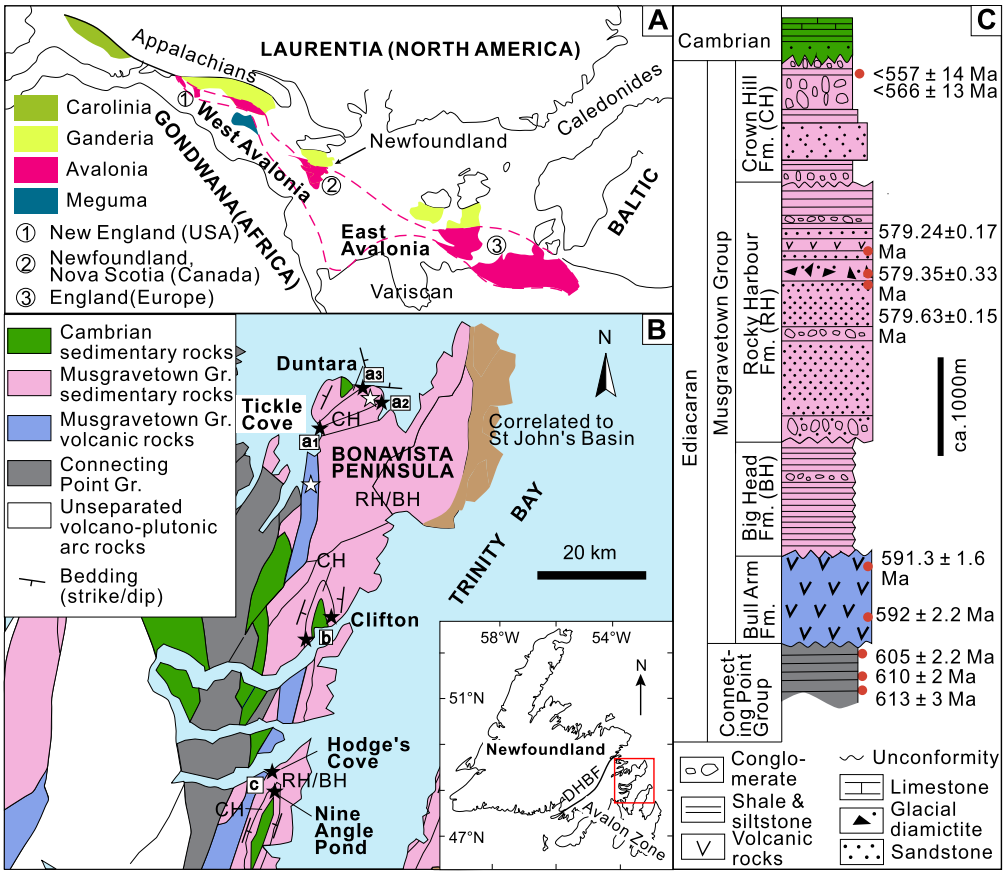


Fig. 1. (A) Location of Avalonia (peri-Gondwana) terranes in a Pangea fit (after Pollock et al., 2009; Henderson et al., 2016; Waldron et al., 2019). (B) Outline of West Avalonia in Newfoundland (inset) and geological map of the study area (modified from O'Brien and King, 2005; Normore, 2010, 2011), showing sampling localities (see detailed stratigraphy and sampling in Fig. 2). Open stars represent the sampling sites in Pisarevsky et al. (2012). (C) Strata of the Musgravetown Group in the study area (after Normore, 2010, 2011) and available age constraints (data are from Pollock et al., 2009; Mills et al., 2016, 2017; Pu et al., 2016). DHBF-Dover-Hermitage Bay Faults. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

tive apparent polar wander path (APWP) segment for that terrane. Integrating the new data with compiled poles from neighboring cratons, we favor an Avalonia-Amazonia juxtaposition that is separated from proto-Gondwana, providing a new interpretation for the Ediacaran–Cambrian paleogeography and assembly of West Gondwana.

2. Geological settings

Avalonia is one of the largest peri-Gondwanan terranes within the Appalachian-Caledonide-Variscan belt, and is subdivided into West Avalonia (in New England and Nova Scotia-Newfoundland, North America) and East Avalonia (in Europe) (Fig. 1A; Henderson et al., 2016; Murphy et al., 2019). Across the entire Avalonia terrane, there is no crystalline basement exposed. Precambrian–early Paleozoic strata from both subunits include early (ca. 760–660 Ma) and main (ca. 635 Ma) stage arc-related rocks, and late stage (<590 Ma) rift-related volcanic and sedimentary successions (Murphy et al., 2008; Henderson et al., 2016). The latter part of the successions hosts Ediacaran Biota and Cambrian–Ordovician Acado-Baltic faunas (O'Brien et al., 1996; Pollock et al., 2009).

The 'Avalon Zone' in southeastern Newfoundland represents the core portion of West Avalonia, and is bounded to the northwest by Ganderia along the Dover-Hermitage Bay Faults (Fig. 1B; O'Brien et al., 1996). Our study area lies in the northwest (Bonavista Peninsula) of the 'Avalonia Zone' (Fig. 1B) and was deposited within the Bonavista Basin according to regional mapping (Normore, 2011). The strata include volcanic arc and deep marine successions of the Love Cove and Connecting Point groups (Mills et al., 2016), and subaerial volcanic and shallow-marine to terrestrial deposits of the overlying Musgravetown Group (Normore, 2010, 2011). In the sampling area, the Musgravetown Group unconformably overlies the >605 Ma Connecting Point Group (Mills et al., 2016) and its top is truncated by the early Cambrian quartzitic Random Formation (O'Brien et al., 1996; Normore, 2010). The total thickness of this group exceeds 4000 m and contains the Bull Arm Formation predominantly bimodal volcanic rocks, Big Head Formation siltstone, Rocky Harbour Formation sandstone, and Crown Hill Formation sandstone/siltstone interbedded with massive conglomerate (Normore, 2010, 2011; Pu et al., 2016). Within the Rocky Harbour Formation, the (ca. 580–579 Ma) Gaskiers glacial diamictites are well preserved (Pu et al., 2016). Together with the updated geochronologic (U-Pb) data (Pollock et al., 2009; Mills et al., 2016, 2017; Pu et al., 2016), the ages for the Musgravetown Group successions are well constrained (Fig. 1C). At its base, the eruption age for the Bull Arm Formation volcanic rocks is constrained to ca. 592 Ma by U-Pb zircon ages from two tuff beds (592 ± 2.2 Ma and 591.3 ± 1.6 Ma; Mills et al., 2017). While at its top, the Crown Hill Formation unconformably overlies the Rocky Harbour Gaskiers-age glacial deposits and is unconformably overlain by the earliest Cambrian Random Formation (Fig. 1C; Pollock et al., 2009; Normore, 2010, 2011; Pu et al., 2016). Considering further the two youngest detrital zircon ages (557 ± 14 Ma and 566 ± 13 Ma) from the most upper portion of this formation (Fig. 1C; Pollock et al., 2009), the age of the Crown Hill Formation is approximately constrained to 560 Ma. Gentle folding during Silurian–Devonian docking of Avalonia to Laurentia (Normore, 2010, 2011) provides opportunity for a paleomagnetic field stability test on the studied rocks.

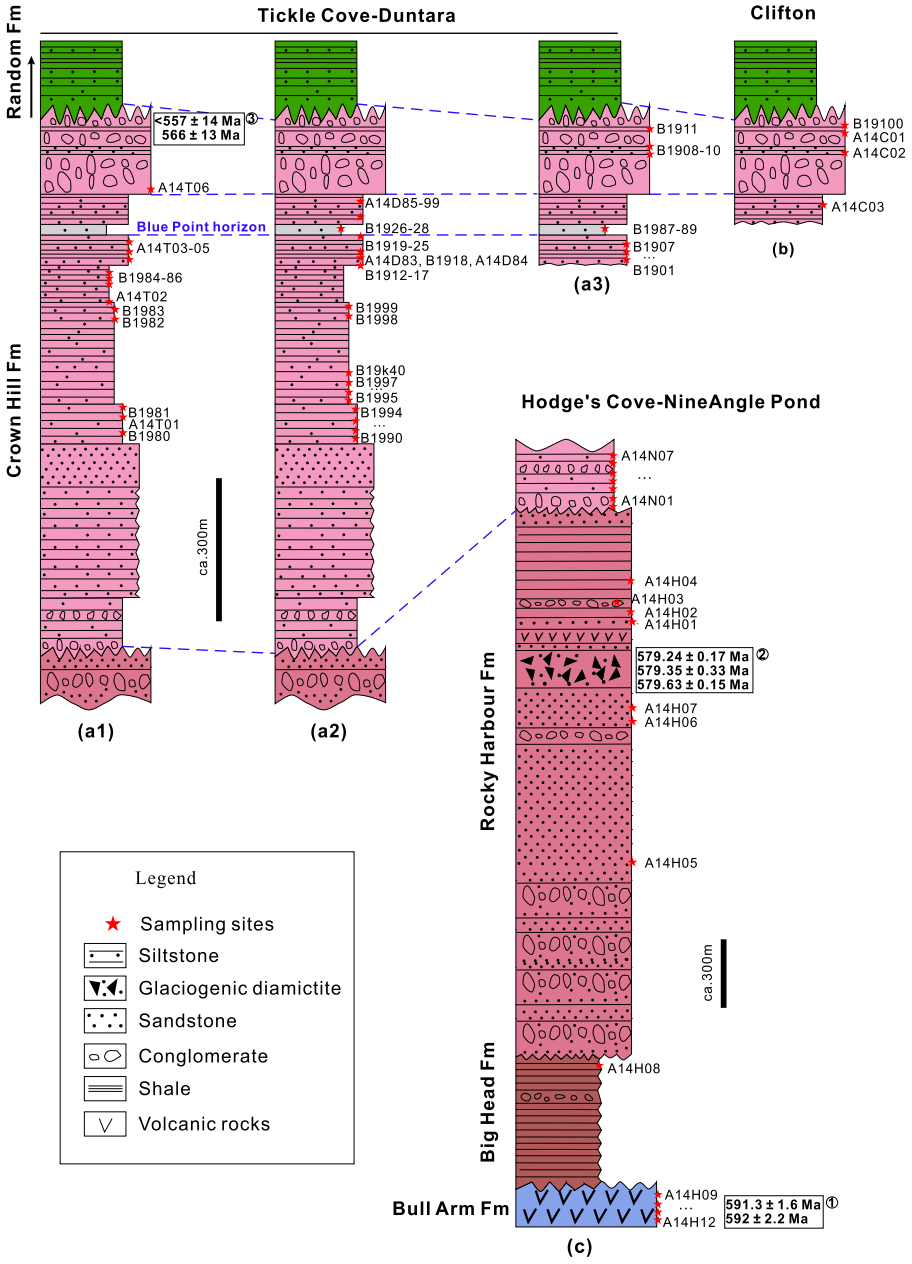


Fig. 2. Stratigraphic sampling of the Musgravetown Group (Bonavista Basin) in this study, showing the available U-Pb zircon ages (① Mills et al., 2016; ② Pu et al., 2016; ③ Pollock et al., 2009).

3. Methods

Samples were collected along five geologic sections at three localities: Tickle Cove-Duntara, Clifton, and Hodge's Cove-Nine Angle Pond (Figs. 1B and 2), encompassing the Bull Arm, Big Head, Rocky Harbour and Crown Hill formations. Most were drilled as oriented cores in the field, and a small number were block-sampled. Magnetic compasses provided sample azimuthal orientation and solar compasses were used whenever possible for the correction of local magnetic declination. Remanent magnetizations were measured using a 2G-Enterprises DC-SQuID magnetometer, which is equipped with an automatic sample-changing system (Kirschvink et al., 2008) in a magnetically shielded laboratory at Yale University. After measuring the natural remanent magnetization (NRM), all samples were thermally demagnetized with an average of ≥16 incremental steps (Fig. 3), until heating to 600 °C (for basalts) and 690 °C (for sandstones/siltstones) or thoroughly demagnetized/unstable. For the basalts, an additional liquid-nitrogen cooling step was made before heating to remove viscous magnetizations of multidomain magnetite (Dunlop and Argyle, 1991). Magnetic remanence directions were defined using principal-component analysis (Kirschvink, 1980). The software packages Paleomag (Jones, 2002) and PaleoMac (Cogné, 2003) were used for data analysis and figure production.

4. Results

Overall, remanence directions from the rocks of the upper portion (Crown Hill Formation) of the Musgravetown Group are more stable and consistent than those from the lower portion. Directions from the Bull Arm and Big Head formations are highly scattered (within site/same formation) or unusable (Table 1; Fig. 3A-C). Only two sites from the Bull Arm Formation yielded the characteristic SW-down and NE-up directions (Fig. 3A and B) as measured previ-

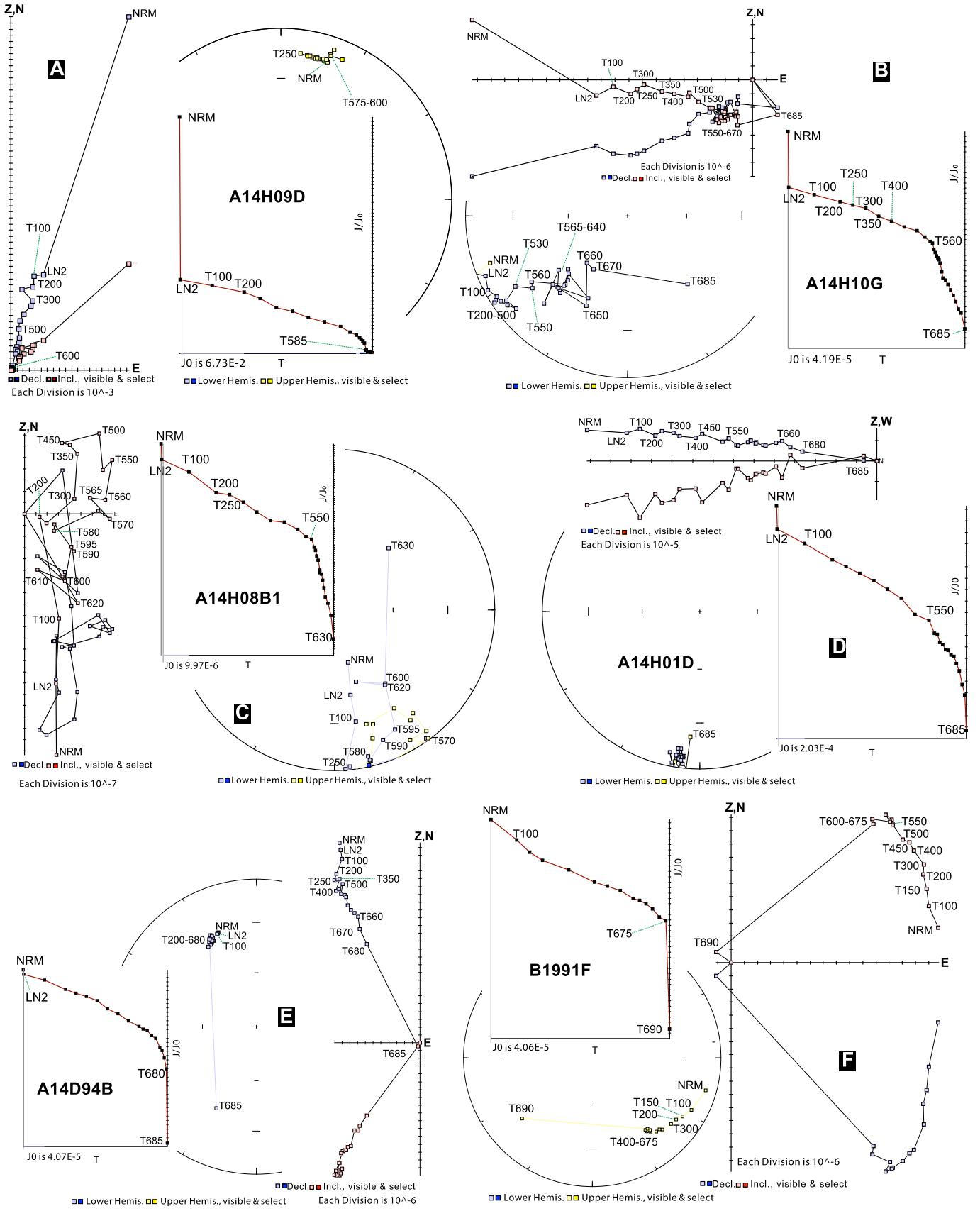


Fig. 3. Progressive demagnetization behaviors for representative specimens from the Bull Arm (A and B), Big Head (C), Rocky Harbour (D) and Crown Hill (E and F) formations. NRM, natural remanent magnetization; LN2, liquid nitrogen cooling step; T100, demagnetization at 100 °C, etc. All projections are in tilt-corrected coordinates.

Table 1
Summary of paleomagnetic results in this study. S-lat/S-long, latitude and longitude for sampling sites; n/n0, number of specimens for mean direction calculation/number of specimens demagnetized; Dg/Ig and Ds/Is, declination and inclination in geographic and stratigraphic coordinates, respectively; A95/α95, radius of 95% confidence circle; MAD, maximum angular deviation; k, precision parameter; PEF, present Earth's field. *Excluded from final calculation; Sites with "a" are from Pisarevsky et al., 2012.

Section	Site	Stratigraphic Unit	S-Lat	S-Long	Strike	Dip	n/n_0	D_g	I_g	D_s	I_s	α_{95}/MAD	k	Interpretation
Clifton	B19100	Crown Hill Fm	48°10'30.70"N	53°38'55.10"W	208	71	6/17	266.2	-66.3	133.7	-38.5	10.3	102.9	Primary
	A14C01	Crown Hill Fm	48°10'39.58"N	53°39'55.08"W	86	24	4/7	144.7	-3.9	141.0	-26.0	9.7	266.3	Primary
	A14C02*	Crown Hill Fm	48°10'39.54"N	53°39'56.05"W	61	33	5/8	122.1	68.1	135.1	37.3	13.3	40.2	Transiton directions?
	A14C03*	Crown Hill Fm	48°10'46.92"N	53°40'19.78"W	46	36	9/9	335.2	-73.2	322.9	-37.9	3.0	255.3	Transiton directions?
Tickle Cove-Duntara	B1911	Crown Hill Fm	48°36'16.40"N	53°23'45.40"W	130	30	9/10	148.7	-38.6	122.7	-41.7	8.8	37.5	Primary
	B1910	Crown Hill Fm	48°36'16.60"N	53°23'36.30"W	195	25	2/8	333.8	57.8	315.7	38.2	15.2	136.0	Primary
	B1909*	Crown Hill Fm	48°36'16.60"N	53°23'36.30"W	195	25	5/6	359.4	69.1	320.5	53.7	21.8	15.6	Dispersed
	B1908*	Crown Hill Fm	48°36'16.60"N	53°23'36.30"W	230	40	0/8	–	–	–	–	–	–	
	A14T06*	Crown Hill Fm	48°35'21.48"N	53°28'49.80"W	46	15	0/8	–	–	–	–	–	–	
	A14D99	Crown Hill Fm	48°35'50.96"N	53°21'19.22"W	275	27	3/3	145.0	-61.2	161.9	-37.8	7.3	190.2	Primary
	A14D98	Crown Hill Fm	48°35'50.57"N	53°21'19.98"W	275	27	4/4	327.3	52.0	339.2	27.8	11.5	48.7	Primary
	A14D97*	Crown Hill Fm	48°35'51.07"N	53°21'20.30"W	283	16	3/3	288.3	-7.7	286.3	-6.4	20.5	24.9	Dispersed
	A14D96	Crown Hill Fm	48°35'51.11"N	53°21'20.92"W	283	16	3/3	318.3	50.1	330.3	39.4	8.1	157.2	Primary
	A14D95	Crown Hill Fm	48°35'51.36"N	53°21'22.57"W	283	16	3/3	136.3	-27.5	142.6	-20.0	7.0	207.1	Primary
	A14D94	Crown Hill Fm	48°35'50.06"N	53°21'26.39"W	274	16	3/3	323.1	47.1	331.4	34.2	2.7	1421.7	Primary
	A14D93	Crown Hill Fm	48°35'49.74"N	53°21'26.60"W	274	16	2/3	331.9	49.6	339.0	35.5	18.4	93.6	Primary
	A14D92	Crown Hill Fm	48°35'49.85"N	53°21'26.93"W	284	18	3/3	309.7	51.5	325.6	41.4	8.7	134.9	Primary
	A14D91	Crown Hill Fm	48°35'49.38"N	53°21'28.73"W	284	18	3/3	144.0	-54.0	157.5	-40.7	10.9	85.5	Primary
	A14D90*	Crown Hill Fm	48°35'49.56"N	53°21'29.23"W	286	14	3/3	332.0	-39.2	321.8	-48.5	16.6	37.5	Anomalous directions?
	A14D89	Crown Hill Fm	48°35'48.80"N	53°21'32.54"W	294	19	3/3	321.5	45.2	334.7	34.4	4.4	536.6	Primary
	A14D88	Crown Hill Fm	48°35'49.16"N	53°21'35.96"W	266	13	4/4	127.0	-41.1	133.9	-31.9	2.7	842.1	Primary
	A14D87	Crown Hill Fm	48°35'49.78"N	53°21'42.77"W	285	16	2/3	133.5	-48.8	146.3	-39.4	8.9	392.3	Primary
	A14D86*	Crown Hill Fm	48°35'49.96"N	53°21'42.84"W	285	16	0/3	–	–	–	–	–	–	
	A14D85	Crown Hill Fm	48°35'48.66"N	53°21'45.22"W	285	16	3/3	119.9	-41.2	131.8	-35.4	20.6	24.6	Primary
	B1987	Crown Hill Fm	48°36'20.21"N	53°21'58.33"W	253	21	8/8	318.4	41.3	323.3	21.8	4.8	119.4	Primary
	B1988	Crown Hill Fm	48°36'18.99"N	53°21'59.06"W	250	26	6/6	317.2	66.7	328.1	41.7	8.0	58.7	Primary
	B1989*	Crown Hill Fm	48°36'16.81"N	53°22'0.02"W	250	26	5/7	139.1	-19.8	140.3	4.6	9.0	187.4	Transiton directions?
	B1928*	Crown Hill Fm	48°35'48.90"N	53°21'45.50"W	249	20	0/3	–	–	–	–	–	–	
	B1927*	Crown Hill Fm	48°35'48.30"N	53°21'46.80"W	250	20	0/4	–	–	–	–	–	–	
	B1926*	Crown Hill Fm	48°35'48.30"N	53°21'46.80"W	250	20	3/3	3.7	60.4	355.3	41.5	35.3	8.8	Dispersed
	B1925*	Crown Hill Fm	48°35'48.30"N	53°21'46.80"W	252	20	5/5	129.3	-9.7	129.5	7.2	15.2	46.0	Transiton directions?
	B1907	Crown Hill Fm	48°36'10.10"N	53°21'57.10"W	322	13	6/10	112.3	-20.3	117.3	-26.2	13.5	33.1	Primary
	B1924	Crown Hill Fm	48°35'48.30"N	53°21'46.80"W	252	20	3/3	101.7	-54.8	119.5	-42.1	4.9	422.4	Primary
	B1923	Crown Hill Fm	48°35'48.30"N	53°21'46.80"W	253	20	4/4	114.9	-49.4	127.0	-34.3	2.5	1037.3	Primary
	B1922	Crown Hill Fm	48°35'48.30"N	53°21'46.80"W	256	20	2/4	100.8	-58.1	122.1	-46.2	8.8	403.7	Primary
	B1921	Crown Hill Fm	48°35'48.30"N	53°21'46.80"W	255	20	4/4	335.1	46.7	337.4	26.9	7.9	102.8	Primary
	B1920	Crown Hill Fm	48°35'48.30"N	53°21'46.80"W	254	20	4/4	329.7	59.5	334.8	39.9	8.5	88.9	Primary
	B1919	Crown Hill Fm	48°35'48.30"N	53°21'46.80"W	255	20	4/4	271.4	85.9	334.1	68.5	7.6	109.2	Primary
	A14D84*	Crown Hill Fm	48°35'48.16"N	53°21'52.70"W	278	13	0/3	–	–	–	–	–	–	
	B1918	Crown Hill Fm	48°35'49.20"N	53°21'53.90"W	279	17	4/4	325.1	61.6	339.5	47.8	9.1	77.6	Primary
	A14D83	Crown Hill Fm	48°35'48.73"N	53°21'53.86"W	275	18	3/3	318.5	50.2	329.7	36.4	3.9	663.2	Primary
	B1917	Crown Hill Fm	48°35'48.90"N	53°21'52.70"W	279	17	4/4	313.1	57.8	329.5	46.3	5.4	221.1	Primary
	B1906*	Crown Hill Fm	48°36'8.40"N	53°21'58.60"W	306	10	2/9	279.7	44.8	289.6	48.5	32.5	30.6	Dispersed
	B1905	Crown Hill Fm	48°36'7.70"N	53°21'59.30"W	306	10	3/9	297.9	36.5	305.4	37.3	6.6	1444.3	Primary
	B1904*	Crown Hill Fm	48°36'7.86"N	53°21'59.41"W	311	16	3/8	29.5	-46.5	23.9	-62.0	31.1	11.2	Dispersed
	B1903*	Crown Hill Fm	48°36'6.30"N	53°22'2.40"W	346	26	0/8	–	–	–	–	–	–	
	B1902*	Crown Hill Fm	48°36'5.60"N	53°22'3.50"W	347	20	5/9	302.5	40.1	319.5	52.1	27.5	21.1	Dispersed

(*continued on next page*)

Table 1 (*continued*)

Section	Site	Stratigraphic Unit	S-Lat	S-Long	Strike	Dip	n/n$_0$	D$_g$	I$_g$	D$_s$	I$_s$	α_{95}/MAD	k	Interpretation
	B1901*	Crown Hill Fm	48°36'4.90"N	53°22'4.60"W	336	32	4/9	304.1	36.7	332.1	45.7	24.3	26.9	Dispersed
	B1916	Crown Hill Fm	48°35'49.20"N	53°21'53.90"W	278	17	4/4	251.5	70.9	303.7	71.5	12.1	43.8	Primary
	B1915*	Crown Hill Fm	48°35'49.10"N	53°21'54.80"W	278	17	0/4	–	–	–	–	–	–	
	B1914	Crown Hill Fm	48°35'48.80"N	53°21'56.50"W	277	21	3/4	290.6	34.7	302.7	27.5	17.2	35.1	Primary
	B1913	Crown Hill Fm	48°35'48.80"N	53°21'56.50"W	276	21	3/3	304.8	43.2	317.9	30.9	9.7	109.2	Primary
	B1912	Crown Hill Fm	48°35'48.80"N	53°21'56.50"W	275	21	4/4	317.4	39.1	326.4	23.8	5.3	228.0	Primary
	A14T05*	Crown Hill Fm	48°33'32.40"N	53°29'25.80"W	315	8	0/8	–	–	–	–	–	–	
	A14T04*	Crown Hill Fm	48°33'23.76"N	53°29'26.52"W	187	34	0/9	–	–	–	–	–	–	
	A14T03	Crown Hill Fm	48°32'57.84"N	53°29'9.60"W	213	26	4/10	161.8	-73.2	139.1	-49.6	18.1	20.1	Primary
	B1986	Crown Hill Fm	48°32'47.30"N	53°28'20.20"W	216	42	7/7	301.4	76.9	304.7	34.9	10.9	27.4	Primary
	B1985	Crown Hill Fm	48°32'46.60"N	53°28'18.40"W	221	43	6/7	295.4	77.1	306.8	34.5	17.7	17.2	Primary
	B1984	Crown Hill Fm	48°32'45.95"N	53°28'17.09"W	233	35	3/7	276.7	73.6	307.0	42.4	13.5	56.5	Primary
	A14T02	Crown Hill Fm	53°27'49.68"W	53°28'12.72"W	199	22	8/8	306.9	61.9	300.0	40.5	5.9	78.8	Primary
	B1983	Crown Hill Fm	48°32'37.70"N	53°28'10.30"W	226	54	5/9	341.8	81.2	320.3	28.0	15.2	21.0	Primary
	B1982	Crown Hill Fm	48°32'37.70"N	53°28'10.30"W	226	54	2/6	333.2	78.5	319.7	25.0	10.3	292.7	Primary
	B1999*	Crown Hill Fm	48°34'46.30"N	53°19'34.70"W	265	20	0/7	–	–	–	–	–	–	
	B1998	Crown Hill Fm	48°34'45.70"N	53°19'32.10"W	260	20	1/6	111.2	-42.6	123.1	-30.4	16.5		Primary
	B1997*	Crown Hill Fm	48°34'36.50"N	53°19'22.30"W	241	21	7/7	359.3	53.5	350.9	34.2	34.0	3.4	Dispersed
	B1996	Crown Hill Fm	48°34'36.60"N	53°19'21.70"W	231	25	8/8	305.4	63.4	312.1	38.9	5.5	90.9	Primary
	B19K40*	Crown Hill Fm	48°34'36.58"N	53°19'20.53"W	213	39	6/8	139.8	-68.7	130.0	-30.3	21.9	23.6	Dispersed
	B1995	Crown Hill Fm	48°34'36.50"N	53°19'20.60"W	213	39	6/8	125.8	-66.6	124.3	-27.6	19.9	18.4	Primary
	B1981	Crown Hill Fm	48°32'29.04"N	53°27'49.68"W	271	15	4/7	161.1	-48.2	165.1	-33.9	15.1	28.6	Primary
	A14T01	Crown Hill Fm	48°32'29.04"N	53°27'49.68"W	285	12	8/9	114.2	-47.4	126.2	-44.2	10.8	29.6	Primary
	B1980*	Crown Hill Fm	48°32'12.90"N	53°27'40.10"W	346	14	7/7	200.9	-70.4	222.4	-60.2	10.8	27.5	?
	B1994	Crown Hill Fm	48°34'36.30"N	53°19'18.40"W	253	23	6/7	140.9	-74.3	153.5	-52.0	6.1	102.0	Primary
	B1993	Crown Hill Fm	48°34'35.52"N	53°19'18.05"W	213	38	5/7	358.7	64.0	328.8	33.8	7.9	75.3	Primary
	B1992	Crown Hill Fm	48°34'34.40"N	53°19'18.80"W	234	32	5/7	294.7	65.9	309.7	35.8	10.8	132.5	Primary
	B1991	Crown Hill Fm	48°34'32.70"N	53°19'21.40"W	250	32	6/6	138.8	-56.0	147.1	-25.3	12.9	42.2	Primary
	B1990	Crown Hill Fm	48°34'32.40"N	53°19'21.30"W	250	32	6/7	332.8	66.2	336.5	34.3	4.6	324.2	Primary
Hodge's	A14N07*	Crown Hill Fm	47°57'26.17"N	53°44'44.88"W	37	63	5/8	271.7	50.8	165.3	54.3	37.0	3.9	Dispersed
Cove-Nine	A14N06*	Crown Hill Fm	47°57'27.22"N	53°44'46.75"W	16	50	5/7	346.8	47.0	45.9	46.2	9.0	59.2	?
Angle Pond	A14N05*	Crown Hill Fm	47°57'28.04"N	53°44'51.76"W	29	55	5/8	8.2	61.3	79.9	38.4	19.5	13.1	PEF
	A14N04*	Crown Hill Fm	47°57'28.80"N	53°44'53.84"W	25	83	0/4	–	–	–	–	–	–	
	A14N03*	Crown Hill Fm	47°57'31.46"N	53°44'53.74"W	30	66	5/7	359.8	78.0	108.5	29.6	12.2	32.1	PEF
	A14N02	Crown Hill Fm	47°57'34.24"N	53°44'57.66"W	31	48	6/7	314.7	-0.2	321.1	45.9	19.9	13.8	Primary
	A14N01*	Crown hill Fm	47°57'47.56"N	53°44'58.81"W	33	71	6/10	14.5	68.2	100.1	24.4	6.5	89.6	PEF
	A14H04*	Rocky Harbour Fm	48° 0'31.75"N	53°42'43.70"W	59-69	60-67	5/12	261.1	32.6	219.9	27.5	20.0	12.4	?
	A14H03	Rocky Harbour Fm; Shale Clasts-for a conglomerate test	48°0'32.26"N	53°42'44.10"W	54	52	A	133.1	55.4	137.8	3.8	9.0		
					54	52	B1	248.5	4.0	239.8	13.9	5.5		
					54	52	B2	156.9	40.8	153.9	-10.3	15.1		
					54	52	C	131.4	-3.6	122.5	-53.7	19.8		
					54	52	D	197.6	-25.9	235.3	-43.6	16.0		
	A14H02*	Rocky Harbour Fm	48° 0'32.72"N	53°42'44.75"W	50	53	0/8	–	–	–	–	–	–	
	A14H01*	Rocky Harbour Fm	48° 0'33.55"N	53°42'45.58"W	41	45	6/9	213.5	42.1	184.6	23.9	30.3	6.5	Dispersed
	A14H07*	Rocky Harbour Fm	48° 0'26.39"N	53°41'33.25"W	224-241	35-40	8/8	49.1	48.1	16.6	34.7	8.0	42.6	?
	A14H06*	Rocky Harbour Fm	48° 0'27.29"N	53°41'31.06"W	205-244	17-45	8/9	52.3	51.0	19.6	35.5	11.5	26.0	?
	A14H05*	Rocky Harbour Fm	48° 0'27.61"N	53°40'15.60"W	236-271	35-43	2/6	56.9	-3.8	64.7	-19.1	18.3	93.9	?
	A14H08*	Big Head Fm	48° 0'48.20"N	53°43'42.60"W	31	15	0/2	–	–	–	–	–	–	
	A14H09	Bull Arm Fm; Basalts	48° 0'48.74"N	53°43'43.82"W	31	15	2/8	19.1	-19.6	14.5	-15.9	3.2		Primary
	A14H10	Bull Arm Fm; Red s.s.	48° 0'32.26"N	53°44'56.26"W	276	8	1/7	217.2	37.3	221.4	44.0	10.4		Primary
	A14H11*	Bull Arm Fm; Basalts	48° 0'32.18"N	53°45'23.15"W	289	25	0/8	–	–	–	–	–	–	

(*continued on next page*)

Table 1 (*continued*)

Section	Site	Stratigraphic Unit	S-Lat	S-Long	Strike	Dip	n/n$_0$	D$_g$	I$_g$	D$_s$	I$_s$	α_{95}/MAD	k	Interpretation
	A14H12*	Bull Arm Fm; Basalts	48° 0'33.05"N	53°45'58.18"W	326	25	7/9	149.3	-28.2	161.6	-24.0	23.0	6.7	Dispersed
	BA1[a]	Bull Arm Fm; Basalts	48° 27.07'N	53°32.48'W	2	66	8/8	207.3	-13.9	205.6	16.5	14.6	15.3	Primary
	BA2*	Bull Arm Fm; Basalts	48° 27.07'N	53°32.48'W	2	66	5/6	99.2	-3.7	111.8	-68.5	24.3	10.9	Dispersed
	BA3[a]	Bull Arm Fm; Basalts	48° 27.07'N	53°32.48'W	2	66	6/7	237.5	9.2	194.7	54.2	14.1	23.5	Primary
	BA4[a]	Bull Arm Fm; Basalts	48° 27.07'N	53°32.48'W	4	76	5/6	211.9	-16.8	208.8	21.6	12.5	36.1	Primary
	BA5*	Bull Arm Fm; Basalts	48° 27.07'N	53°32.48'W	4	76	6/6	182.1	-24.6	207.6	-6.0	7.4	82.9	?
	BA6[a]	Bull Arm Fm; Basalts	48° 23.65'N	53°33.33'W	349-350	45-60	3/6	52.6	-13.0	50.3	-28.4	13.7	81.8	Primary
	BA8[a]	Bull Arm Fm; Basalts	48° 27.52'N	53°31.68'W	4	76	5/8	59.2	17.5	54.0	-42.8	14.8	27.9	Primary
	BA10[a]	Bull Arm Fm; Basalts	48° 27.77'N	53°31.67'W	354	64	3/6	52.0	-8.3	18.4	-54.9	14.2	76.5	Primary
	BA11[a]	Bull Arm Fm; Basalts	48° 27.77'N	53°31.67'W	354	64	3/6	62.9	-15.6	17.9	-67.7	11.7	111.9	Primary
	BA12[a]	Bull Arm Fm; Basalts	48° 27.77'N	53°31.67'W	354	64	5/6	209.3	-10.5	201.2	25.5	3.4	507.2	Primary
Mean$_{CH}$		**Crown Hill Fm.**			**51 Sites**			**313.6(133.6)**	**55.7(-55.7)**			**5.4**	**14.4**	
										320.2(140.2)	**37.7(-37.7)**	**3.6**	**31.1**	
												$A_{95} = 3.3$		
Paleopole$_{CH}$					**48.1 (-48.1) N, 189.9 (9.9) E; dm/dp=4.2/2.5**									
Mean$_{BA}$		**Bull Arm Fm**			**10 Sites**			**223.1(43.1)**	**4.5(-4.5)**			**14.8**	**11.6**	
										209.3(29.3)	**37.9(-37.9)**	**13.1**	**14.5**	
												$A_{95} = 11.9$		
Paleopole$_{BA}$					**-15.5 (15.5) N, 278.2 (98.2) E; dm/dp=15.5/9.1**									

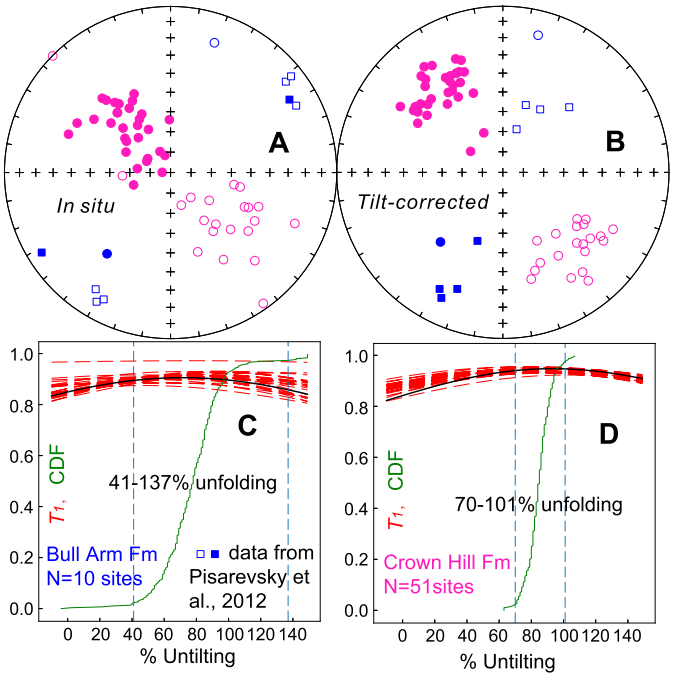


Fig. 4. Paleomagnetic results of the Bull Arm (blue; together with data from Pisarevsky et al., 2012) and Crown Hill formations (pink). (A–B) Equal-area stereographic projections for the directions in *in-situ* and tilt-corrected coordinates, respectively. Solid/open symbols represent the lower/upper hemisphere projections. (C–D) Parametric bootstrap fold tests (Tauxe et al., 2016) for the ChRMs of Bull Arm and Crown Hill formations. T_1/CDF, maximum eigenvalue/cumulative distribution function (CDF) of the percent unfolding.

ously from the Bonavista Basin (Fig. 1B) by Pisarevsky et al. (2012). Put together, the maximum concentration of the characteristic remanent magnetizations (ChRMs) occurs in the range 41–137% unfolding at the 95% confidence level (Fig. 4C; Tauxe et al., 2016). The result for this test is probably biased by the low number of sites applied, but three positive conglomerate tests and a baked contact test (Pisarevsky et al., 2012) in this region strongly support a primary origin for the remanence. Thus, a combined paleomagnetic pole ('BA-b'; Tables 1 and 2) for the ca. 592 Ma Bull Arm Formation at Bonavista is calculated at $\lambda = -15.5°$N, $\varphi = 278.2°$E, $A_{95} = 11.9°$ with a paleolatitude of $21.3 \pm 9.1°$. An indistinguishable pole relying on the dataset of Pisarevsky et al. (2012) is also calculated ($\lambda = -14.0°$N, $\varphi = 277.9°$E, $A_{95} = 14.2°$) for comparison. For the Rocky Harbour Formation, ChRMs are inconsistent between different sites (Table 1). An inconclusive conglomerate test (Fig. 5) suggests that their reliabilities are questionable and are excluded from further discussion.

In contrast, samples from the overlying Crown Hill Formation shared broadly similar characteristics to each other, exhibiting one single (Fig. 3E) or two magnetic components (Fig. 3F): low-(below 300–500 °C; LTC) and high-temperature (up to 650–690 °C hematite range; HTC) components. The LTC directions are parallel to the present Earth's field (PEF; Fig. 6A) and become scattered after tilt correction (Fig. 6B), representing a viscous magnetization of recent field. Both the HTCs and single stable components linearly decay to the origin (Fig. 3E and F) or rarely along a demagnetization plane through the origin, defining the ChRMs which show dipolar-intermediate NW-down/SW-up directions (Fig. 4A and B). The ChRMs pass a parametric bootstrap fold test (Fig. 4D; Tauxe et al., 2016) and a reversal test at the 95% confidence level (class 'B'; McFadden and McElhinny, 1990). An average site-mean direction for the combined polarities is calculated at $D_g = 133.6°$, $I_g = -55.7°$, $k_g = 14.4$, $\alpha_{95} = 5.4°$; $D_s = 140.2°$, $I_s = -37.7°$,

Table 2
Ediacaran paleomagnetic poles of Avalonia. Notes: P-lat/P-long, latitude/longitude of paleomagnetic pole. A95, the radius of the paleomagnetic pole 95% confidence limits. Q_f, Q-factor of Van der Voo, 1990. "*" represents poles not used in Fig. 7.

Pole ID	Rock Unit (Area)	Age (Ma)	Age range (Ma)		P-lat (°N)	P-long (°E)	A_{95} (°)	1	2	3	4	5	6	7	Q_f	Reference
			Max	Min												
West Avalonia																
CH	**Crown Hill Fm, Newfoundland, Canada**	**< 557+/-14 Ma; 566+/-13**	**570**	**540**	**-48.1**	**9.9**	**3.3**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**7**	**This study; ages from Pollock et al., 2009**
BA-b	**Bonavista Bull Arm Fm, Newfoundland, Canada**	**592 ± 2.2; 591.3 ± 1.6**	**594**	**589**	**-15.5**	**278.2**	**11.9**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**7**	**This study; Pisarevsky et al., 2012; ages from Mills et al., 2017**
BA-a	Argentia Bull Arm Formation, Newfoundland, Canada	< 605			-2.9	256.3	10.6	0	1	1	1	1	1	1	6	Pisarevsky et al., 2012; reference age of Mills et al., 2017
MGv	Marystown Group volcanic-sedimentary sequence, Newfoundland, Canada	585 ± 2; 580 ± 3; 576.8 ± 2.6; 575 ± 2	587	573	26.6	239.6	11.1	1	1	1	1	1	1	1	7	McNamara et al., 2001; Sparkes and Dunning, 2014
Rf*	Roxbury Formation, New England, USA	~ 582 Ma, correlation to Gaskiers (Pu et al., 2016)	593	570	13	267	8	0	1	1	0	0	0	1	3	Wu et al., 1986; age from Thompson et al., 2007, 2014
LMv	Lynn-Mattapan Volcanics, New England, USA	595.8 ± 1.2; 597.4 ± 1.5; 596.0 ± 1.4; 595.7 ± 1.6	599	594	60	206	8.4	1	1	1	1	1	1	1	7	Thompson et al., 2007
East Avalonia																
Cv	Caldecote Volcanics, England	603 ± 2	605	601	-5	329.7	8.2	1	1	1	1	1	0	1	6	Vizan et al., 2003

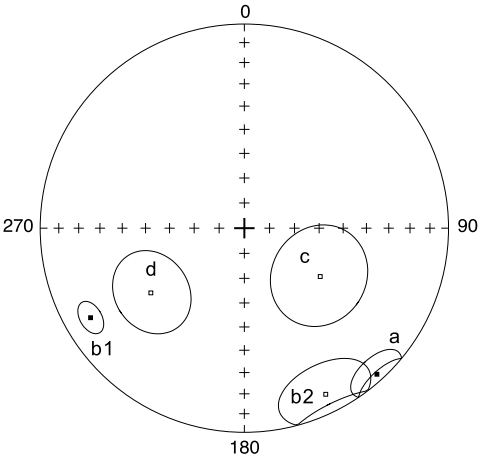


Fig. 5. Distribution of ChRMs from the red sandstone pebbles of Rocky Harbour Formation, showing an inconclusive conglomerate test. b1 and b2 are drilled from the same clast. Solid/open squares represent the lower/upper hemisphere directions.

$k_s = 31.1$, $\alpha_{95} = 3.6°$ (n = 51 sites), yielding a new paleopole ('CH') at $\lambda = -48.1°$N, $\varphi = 09.9°$E, $A_{95} = 3.3°$.

5. Discussion

5.1. Apparent polar wander path of Avalonia and its implications

Our results largely reproduce those presented by Pisarevsky et al. (2012) from the same formations, but we augment their dataset with Crown Hill Formation sites that lie within the typical northeasterly tectonic grain of the Bonavista Peninsula (Fig. 1B). Thus, we are able to show that the Bull Arm versus Crown Hill remanences' azimuthal directional discordance is not due to local rotations; the Newfoundland sector of West Avalonia can be considered as a quasi-rigid block for the purposes of terrane reconstructions. Integrating our new results with the compiled paleopoles (with Q-factor >3; Van der Voo, 1990) from West Avalonia of Newfoundland (Table 2), a segment of APWP for this terrane is defined, showing a large (~90°) polar motion from ca. 590–575 to 560 Ma (Fig. 7A). A reconstructed Avalonia composite terrane is also resolved by fitting the Ediacaran poles from New England, Newfoundland, and England (Fig. 7A; Table 2).

The substantial arc distance between mid- and late-Ediacaran poles (~90° over a timespan of merely ~20–30 Myr) from Avalonia is reminiscent of the comparable distances within APWPs of both Laurentia (McCausland et al., 2011; Mitchell et al., 2011) and West Africa (Robert et al., 2017, 2018) over the same time interval. To explain these kinds of variant datasets, Abrajevitch and Van der Voo (2010) proposed an equatorial geomagnetic dipole hypothesis, whereas Robert et al. (2017, 2018) interpreted them as recording inertial interchange true polar wander (IITPW). Although the contribution of an equatorial dipole field during the polarity reversals is permissible (Halls et al., 2015), the available data overall weighs more towards the IITPW process. As pointed out by many researchers (e.g., McCausland et al., 2007, 2011; Bono and Tarduno, 2015; Robert et al., 2017, 2018), the multiple-component (steep and shallow) directions from a same rock unit are not equally reliable (reviewed in Robert et al., 2017). Adopting that methodology of data selection based purely on reliability rather than *a priori* judgment of tectonic implications, the most robust poles from Laurentia and West Africa (Table S1; Robert et al., 2017, 2018) define a similar trend of APWPs from ca. 590–575 to 560 Ma (Fig. 7B) and a polar-motion rate is calculated at ~2–9°/Ma (including this study). This rate is consistent with a plausible speed of TPW (e.g., Steinberger and Torsvik, 2008; Greff-Lefftz and Besse, 2014; Robert et al., 2018). Additionally, the quasi-static Sutton hotspot under Laurentia suggests stability of that plate relative to a deep mantle reference frame, which can be adequately explained by TPW in this interval (Mitchell et al., 2011).

After rotating Avalonia to the West African reference frame around an Euler pole (5.8°S, 158.8°E, -84.1°) to fit its two groups (~590–575 and 560 Ma) of poles with those of West Africa and Laurentia (Table S1; Robert et al., 2017, 2018), a composite alignment of APWPs for the continents of West Gondwana is achieved, and a large distance between Avalonia and West Africa is implied (Fig. 7B). The effects of possible inclination shallowing (using a nominal value of $f = 0.6$; dashed 'CH' ellipse in Fig. 7B) are minor at this scale of APWP. Due to ambiguous geomagnetic polarity and ~90° separation between the poles of Avalonia, we consider four possible relative positions of Avalonia and West Africa (positions 1-4; Fig. 7B), none of which allow a direct Avalonia-West Africa juxtaposition. Among the four possibilities, position #1 seems the most reasonable because: (i) it readily accommodates a transition toward West African sources of detritus for early Cambrian Avalonian sediments (Waldron et al., 2019), and (ii) it provides the most feasible way for Avalonia to drift from southern-hemisphere Gondwana (e.g., Torsvik et al., 2012) to eventual accretion with Laurussia in a right-way-up orientation at moderate southerly latitudes during the mid-Paleozoic time (Murphy et al., 2019).

5.2. Paleogeography of West Gondwana

It is well documented that Avalonia had accreted to a larger cratonic margin by ca. 650–630 Ma (e.g., Murphy et al., 2000, 2019), as evidenced by localized high-grade metamorphism in some parts

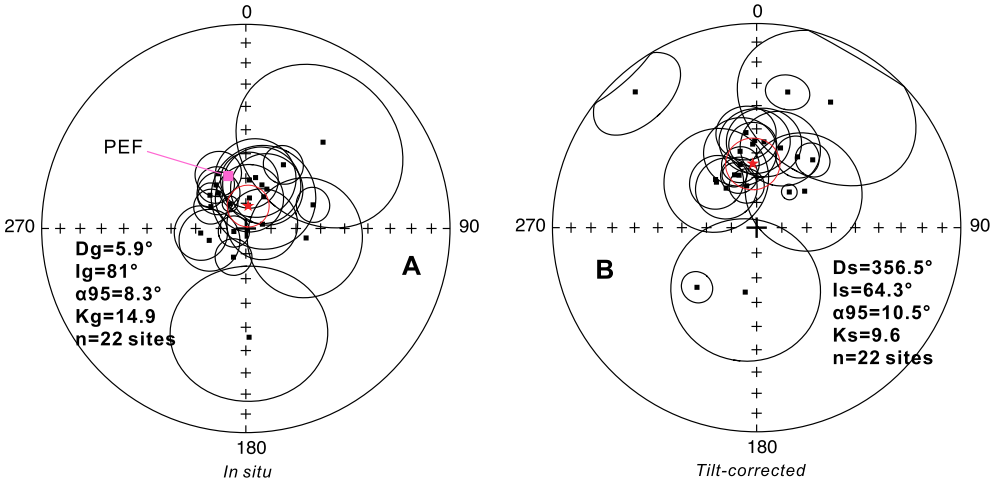


Fig. 6. Directions of the low temperature components (LTCs) for samples from the Crown Hill Formation. Dg/Ds, Ig/Is, and Kg/Ks, declination, inclination and precision parameter for the directions in geographic (A) and stratigraphic (B) coordinates, respectively. Mean directions are shown with stars and $\alpha 95$ confidence ovals. PEF, present Earth's field.

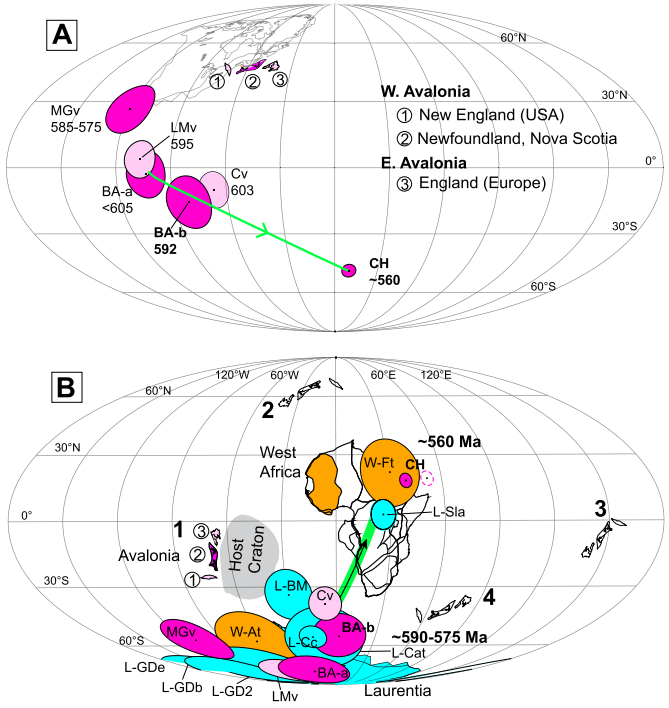


Fig. 7. (A) Apparent polar wander path (APWP) of Avalonia terrane, showing the large polar motion from ca. 590-575 to 560 Ma. See pole list and abbreviations in Table 2. Avalonia composite terrane is restored by rotating East Avalonia (79.1°N, -163.0°E, -34.8°) and New England (43.1°N, -73.6°E, 92.8°) to Newfoundland. The dispersion of Newfoundland poles can be explained by modest amounts of local vertical-axis rotations between the sampled regions, and/or minor APW within the 600-575 Ma interval. (B) Fitting the mid–late Ediacaran APWP segment of Avalonia with the coeval paleomagnetic poles (Table S1) from West Africa and Laurentia after Robert et al. (2017), showing the four possible positions of Avalonia (Euler pole for preferred position #1, Newfoundland to West Africa: 5.8°S, 158.8°E, -84.1°). Dashed-line and unshaded version for pole CH includes correction for a plausible amount of inclination shallowing ($f = 0.6$).

of Avalonia (e.g., O'Brien et al., 1996; Murphy et al., 2004) and an isotopic shift to more evolved crustal sources (Henderson et al., 2016). One of the previously suggested options for that connection is a location ('X' in Fig. 8A) near West Africa, Amazonia, and the southern margin of Baltica (Thompson et al., 2012). By this conventional interpretation, aligning the mid–late Ediacaran APWP segments of Baltica and West Africa (Fig. 8A) creates a gap that represents the most suitable position for Amazonia (Amazonia itself lacks reliable late Ediacaran paleomagnetic data; Fig. 8B). This model would resemble the pre-600 Ma assembly scenario for West Gondwana (e.g., Cordani et al., 2013); however, the location ('X') of Avalonia in this model is not permitted by our Crown Hill data at ~560 Ma, which instead would favor a late Ediacaran location on the Timanian side of Baltica (Fig. 8A and B). But in order for Avalonia to achieve its early Cambrian position near 'X' that is indicated by biogeography (e.g. Cocks and Fortey, 2009; Álvaro et al., 2013) and sedimentary provenance (Pollock et al., 2009; Henderson et al., 2016; Waldron et al., 2019), Avalonia would need to migrate around Baltica in a complicated terminal Ediacaran kinematic scenario, prior to Paleozoic separation and crossing of the Iapetus realm (e.g., Murphy et al., 2004, 2019; Pollock et al., 2009, 2012).

As an alternative model, we place Amazonia as the host craton adjacent to Avalonia—satisfying both basement isotopic similarities (Nance and Murphy, 1994) and sedimentary provenance (Pollock et al., 2009; Henderson et al., 2016). Baltica's position, due to the uncertainty on the ages of the B-Kg and B-Bk (Kurgashlya and Bakeevo formations) paleopoles (Table S1; Robert et al., 2017), is adjusted to retain a loose connection with Avalonia (Thompson et al., 2012) and a broad APWP match with that of other cratons (Fig. 8C). The common potential source of 1.6–1.0 Ga zircons from both Baltica and Amazonia (Pollock et al., 2009; Henderson et al., 2016; Murphy et al., 2019) and a long-lived Meso- to early-Neoproterozoic connection with Laurentia (e.g., Pisarevsky et al., 2008) may provide some support for this juxtaposition. The ~635–570 Ma large strike-slip movements along Avalonia's northern margin (current coordinates) (Murphy et al., 2004; van Staal et al., 2020) may have been caused by the separation of Baltica after its initial rifting from Laurentia at ca. 615 Ma during the opening of Iapetus Ocean (e.g., Pisarevsky et al., 2008). In this context, Amazonia becomes the only feasible host craton to Avalonia and the resulting configuration of West Gondwana at ca. 590–575 Ma in an absolute (TPW-based) reference frame is attained (Fig. 8D). In this model, subduction along the outboard active margin of Avalonia likely resulted in the development of the ca. 560–550 Ma blueschists in Anglesey-Lleyn of Wales, UK (Kawai et al., 2007). Moreover, this new model implies a ~3000 km wide Clymene Ocean between Avalonia-Amazonia and West Africa, compatible with the Ediacaran–Cambrian lithostratigraphic disparities between Avalonia and West Africa (e.g., Landing, 1996, 2005; Landing et al., 2013). Subsequently, West Gondwana was completely assembled by Cambrian time (Trindade et al., 2006; To-

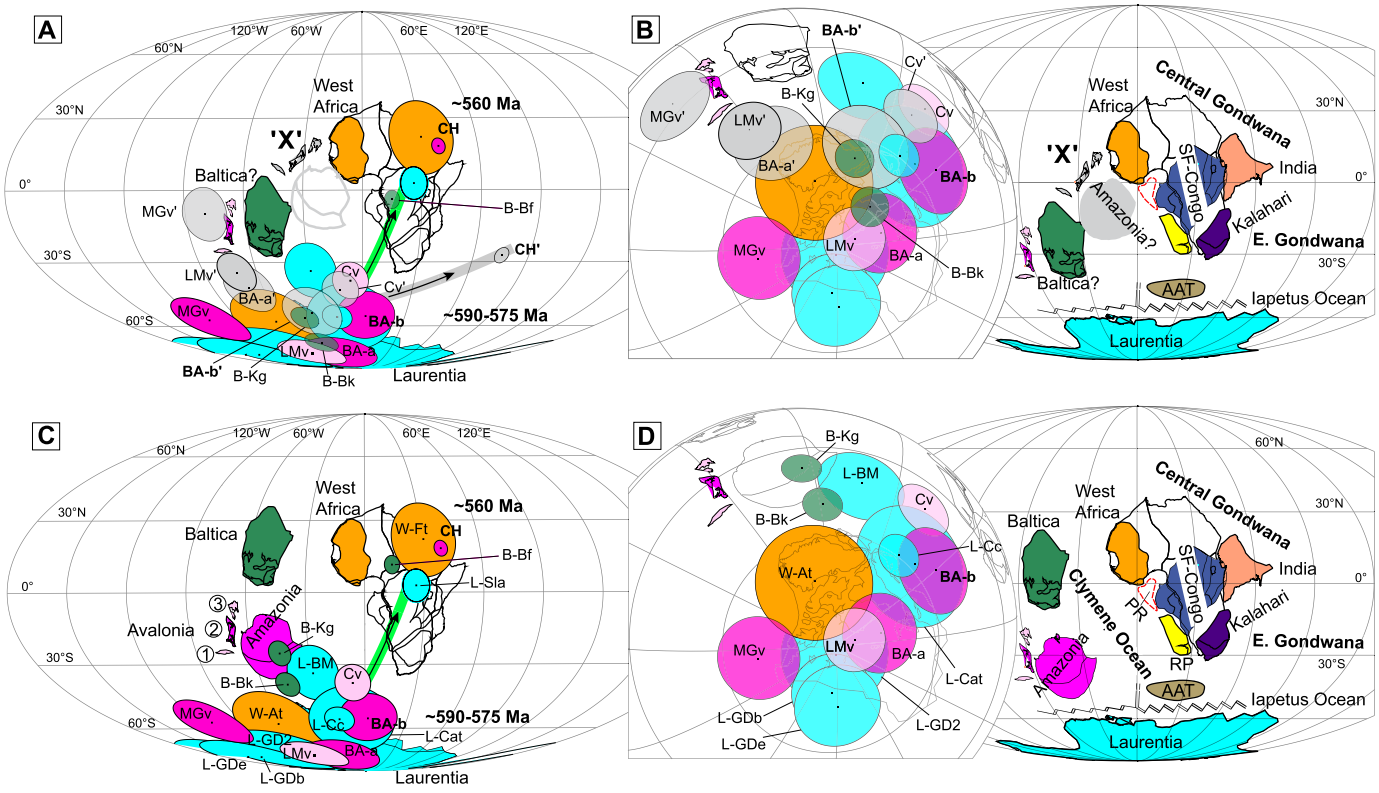


Fig. 8. (A) The mid-late Ediacaran APWP fits (in West African coordinates) and (B) reconstruction of West Gondwana at ca. 590-575 Ma in an absolute reference frame with a Baltica-Avalonia juxtaposition, showing the oft-suggested but paleomagnetically invalid 'X' position of Avalonia (poles shown in gray color; Euler rotation: 41.9°N, 12.3°E, 54.5°). Baltica is rotated to West Africa using Euler rotation (0.2°N, -169.0°E, -117.9°). The poles from Laurentia are rotated to West Africa after Robert et al. (2017). Pole list and abbreviations are in Tables 2 and S1. (C and D) As in (A) and (B), but with Amazonia (12.9°S, 36.7°E, 327.1°) juxtaposed to Avalonia and showing the optimal match for Baltica (see the main text; Euler rotation: 8.4°N, 3.5°E, 92.7°). SF = São Francisco; RP = Río de La Plata; PR = Parnaíba (after Schmitt et al., 2018); AAT = Arequipa-Antofalla Terrane (after Escayola et al., 2011).

hver et al., 2010, 2012), and Avalonia migrated with Amazonia to arrive in proximity to West Africa as a result of that accretion. This amalgamation is consistent with the early Cambrian faunal similarities (such as trilobite genera) between Avalonia and the Western Mediterranean (e.g., Cocks and Fortey, 2009; Álvaro et al., 2013), and the collision could be represented by the early Cambrian Araguaia-Paraguai-Pampean belt (e.g., Tohver et al., 2010; Schmitt et al., 2018) related to Clymene Ocean closure (Fig. 8D). It is worth noting that, before that convergence, the Parnaíba (PR) and Río de La Plata (RP) blocks had been welded together with Central Gondwana by ca. 620–580 Ma (Fig. 8D; reviewed in Schmitt et al., 2018), corresponding to the earlier closure of the Goiás-Pharusian Ocean (or Pan-African-Brasiliano Orogens; Cordani et al., 2013; Ganade de Araújo et al., 2014; Schmitt et al., 2018). This scenario suggests that these paleocontinents witnessed both the early- and late-stage collisions within West Gondwana. Within that tectonic context, our data and model for mid–late Ediacaran time provide a key snapshot onto the evolving Rodinia-Gondwana supercontinental transition, after final Rodinia breakup (Laurentia separating from Baltica and Amazonia) and prior to Gondwana amalgamation. A wide Clymene Ocean persisting after Iapetus opening negates the existence of the Pannotia supercontinent according to classic paleogeographic idealizations of that landmass (e.g., Dalziel, 1997).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

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